ExamGOAL
PYQ PDF

Free · With answer key

JEE MAIN · QUESTION PAPER

JEE Main 2026 (Online) 23rd January Evening Shift

JEE Main Question Paper with Answer Key

23 Jan 2026	75	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

Chemistry	25 questions
Mathematics	25 questions
Physics	25 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Chemistry 25 questions

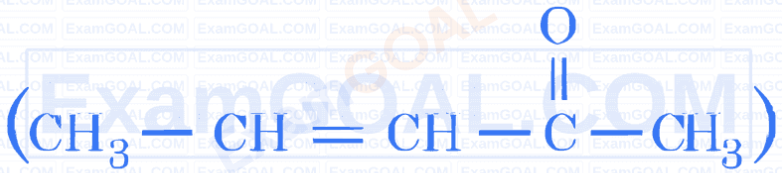
Q1. MCQ (Single Correct Answer)

Which of the following statements are TRUE about Haloform reaction?

A. Sodium hypochlorite reacts with KI to give KOI .

B. KOI is a reducing agent.

C. α, β-unsaturated methylketone



will give iodoform reaction.

D. Isopropyl alcohol will not give iodoform test.

E. Methanoic acid will give positive iodoform test.

Choose the correct answer from the options given below :

(A) A, B & C Only (B) A, C & E Only (C) B, D & E Only (D) A & C Only

Open solution

Q2. MCQ (Single Correct Answer)

The oxidation state of chromium in the final product formed in the reaction between KI and acidified $K_2Cr_2O_7$ solution is :

(A) +2 (B) +3 (C) +4 (D) +6

Open solution

Q3. MCQ (Single Correct Answer)

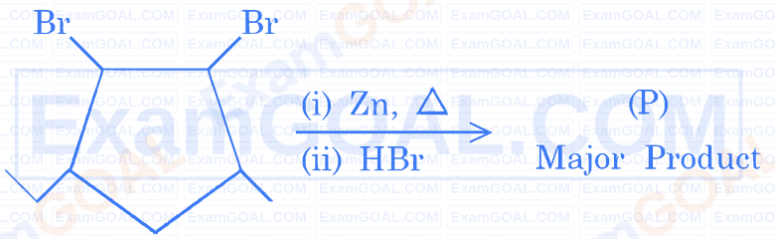
In Carius method 0.2425 g of an organic compound gave 0.5253 g silver chloride. The percentage of chlorine in the organic compound is

(A) 87.65% (B) 53.58% (C) 37.57% (D) 34.79%

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

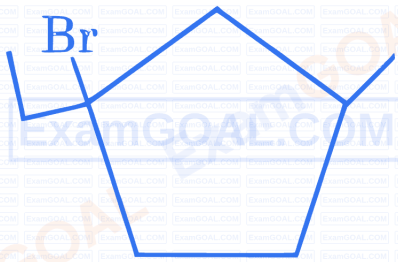
Q4. MCQ (Single Correct Answer)



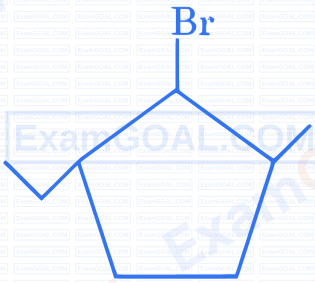
Open solution

Identify (P)

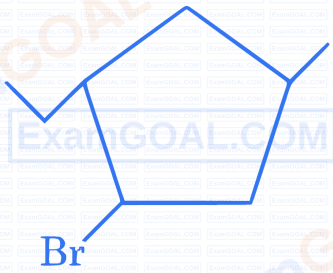
(A)



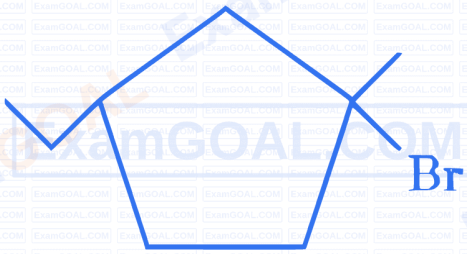
(B)



(C)



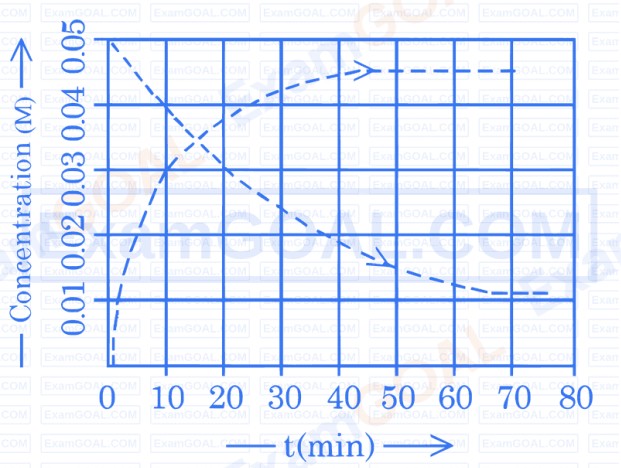
(D)



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q5. MCQ (Single Correct Answer)

Open solution



Given above is the concentration vs time plot for a dissociation reaction : $\mathrm{A} \rightarrow \mathrm{nB}$.

Based on the data of the initial phase of the reaction (initial 10 min), the value of n is ____ .

(A) 2 (B) 5 (C) 4 (D) 3

Q6. MCQ (Single Correct Answer)

Open solution

Observe the following reactions at $\mathrm{T(K)}$.

I. $\mathrm{A} \rightarrow$ products.

II. $5\mathrm{Br}^{-}(\mathrm{aq}) + \mathrm{BrO_3}^{-}(\mathrm{aq}) + 6\mathrm{H}^{+}(\mathrm{aq}) \rightarrow 3\mathrm{Br_2}(\mathrm{aq}) + 3\mathrm{H_2O}(\mathrm{l})$

Both the reactions are started at 10.00 am . The rates of these reactions at 10.10 am are same. The value of $-\frac{\Delta[\mathrm{Br}^{-}]}{\Delta \mathrm{t}}$ at 10.10 am is $2 \times 10^{-4}\ \mathrm{mol\,L^{-1}\,min^{-1}}$. The concentration of A at 10.10 am is $10^{-2}\ \mathrm{mol\,L^{-1}}$. What is the first order rate constant (in min^{-1}) of reaction I ?

(A) 4×10^{-3} (B) 2×10^{-3} (C) 10^{-3} (D) 10^{-2}

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q7. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I:  can be synthesized from using simpler reagents in the order i) Acidic $KMnO_4$, ii) Ammonia, iii) Bromine and alkali

Statement II:  can be converted into using reagents in the order i) Bromine-H_2O ii) $NaNO_2$/HCl (0 - 5 °C) (iii) Aq. H_3PO_2 .

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Q8. MCQ (Single Correct Answer)

Open solution

The work functions of two metals $(M_A$ and $M_B)$ are in the $1:2$ ratio. When these metals are exposed to photons of energy 6 eV , the kinetic energy of liberated electrons of $M_A : M_B$ is in the ratio of $2.642:1$. The work functions (in eV) of M_A and M_B are respectively.

(A) $3.1, 6.2$ (B) $1.5, 3.0$ (C) $2.3, 4.6$ (D) $1.4, 2.8$

Q9. MCQ (Single Correct Answer)

Open solution

Iodoform test can differentiate between

A. Methanol and Ethanol

B. CH_3COOH and CH_3CH_2COOH

C. Cyclohexene and cyclohexanone

D. Diethyl ether and Pentan-3-one

E. Anisole and acetone

Choose the **correct** answer from the options given below :

(A) A, B & E Only (B) A & D Only (C) A & E Only (D) B, C & E Only

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q10. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : $(CH_3)_3\overset{\oplus}{C}$ is more stable than $\overset{\oplus}{C}H_3$ as nine hyperconjugation interactions are possible in $(CH_3)_3\overset{\oplus}{C}$.

Statement II : $\overset{\oplus}{C}H_3$ is less stable than $(CH_3)_3\overset{\oplus}{C}$ as only three hyperconjugation interactions are possible in $\overset{\oplus}{C}H_3$.

In the light of the above statements, choose the **correct** answer from the options given below

(A) Statement I is false but Statement II is true

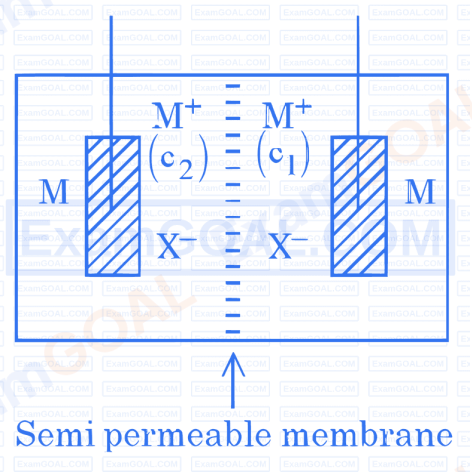
(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q11. MCQ (Single Correct Answer)

Open solution



Consider the above electrochemical cell where a metal electrode (M) is undergoing redox reaction by forming M^+ ($M \rightarrow M^+ + e^-$). The cation $M^+$is present in two different concentrations c_1 and c_2 as shown above. Which of the following statement is correct for generating a positive cell potential?

(A) If c_1 is present at anode, then $c_1 > c_2$.

(B) If c_1 is present at cathode, then $c_1 > c_2$.

(C) If c_1 is present at cathode, then $c_1 < c_2$.

(D) If c_1 is present at anode, then $c_1 = c_2$.

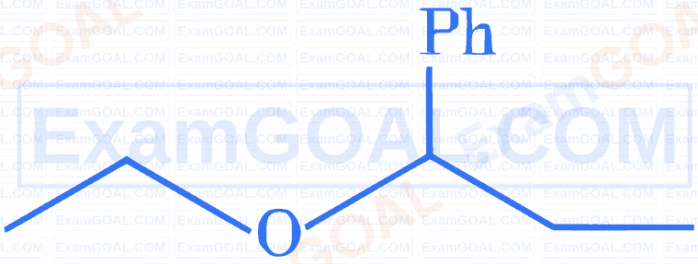
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q12. MCQ (Single Correct Answer)

Open solution

A mixed ether (P), when heated with excess of hot concentrated hydrogen iodide produces two different alkyl iodides which when treated with aq. NaOH give compounds (Q) and (R). Both (Q) and (R) give yellow precipitate with NaOI . Identify the mixed ether (P) :

(A)



(B)



(C)



(D)



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q13. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : The second ionisation enthalpy of Na is larger than the corresponding ionisation enthalpy of Mg .

Statement II : The ionic radius of O^{2-} is larger than that of F^{-}.

In the light of the above statements, choose the ***correct*** answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q14. MCQ (Single Correct Answer)

Open solution

Both human DNA and RNA are chiral molecules. The chirality in DNA and RNA arises due to the presence of

(A) L-sugar component

(B) D-sugar component

(C) Chiral phosphate unit

(D) Base unit

Q15. MCQ (Single Correct Answer)

Open solution

A student has been given a compound " x " of molecular formula- C_6H_7N. ' x ' is sparingly soluble in water. However, on addition of dilute mineral acid, ' x ' becomes soluble in water. ' x ' when treated with $CHCl_3$ and $KOH(alc)$, ' y ' is produced. ' y ' has a specific unpleasant smell. On treatment with benzenesulphonyl chloride, ' x ' gives a compound ' z ' which is soluble in alkali. The number of different " H " atoms present in ' z ' is:-

(A) 8

(B) 4

(C) 5

(D) 7

Q16. MCQ (Single Correct Answer)

Open solution

Identify the **INCORRECT** statements from the following :

A. Notation $^{24}_{12}Mg$ represents 24 protons and 12 neutrons.

B. Wavelength of a radiation of frequency $4.5 \times 10^{15}\ s^{-1}$ is $6.7 \times 10^{-8}\ m$.

C. One radiation has wavelength $= \lambda_1(900\ nm)$ and energy $= E_1$. Other radiation has wavelength $= \lambda_2(300\ nm)$ and energy $= E_2$. $E_1 : E_2 = 3 : 1$.

D. Number of photons of light of wavelength 2000 pm that provides 1 J of energy is 1.006×10^{16}.

Choose the correct answer from the options given below :

(A) A and D Only

(B) A and C Only

(C) A and B Only

(D) B and C Only

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q17. MCQ (Single Correct Answer)

Open solution

Which statements are **NOT TRUE** about XeO_2F_2?

A. It has a see-saw shape.

B. Xe has 5 electron pairs in its valence shell in XeO_2F_2.

C. The $O-Xe-O$ bond angle is close to 180°.

D. The $F-Xe-F$ bond angle is close to 180°.

E. Xe has 16 valence electrons in XeO_2F_2.

Choose the ***correct*** answer from the options given below :

(A) B, D and E Only **(B)** B and D Only **(C)** B, C and E Only **(D)** A and D Only

Q18. MCQ (Single Correct Answer)

Open solution

It is noticed that Pb^{2+} is more stable than Pb^{4+} but Sn^{2+} is less stable than Sn^{4+}. Observe the following reactions.

$PbO_2 + Pb \rightarrow 2PbO; \Delta_rG^\circ(1)$

$SnO_2 + Sn \rightarrow 2SnO; \Delta_rG^\circ(2)$

Identify the correct set from the following

(A) $\Delta_rG^\circ(1) > 0; \Delta_rG^\circ(2) < 0$

(B) $\Delta_rG^\circ(1) < 0; \Delta_rG^\circ(2) > 0$

(C) $\Delta_rG^\circ(1) < 0; \Delta_rG^\circ(2) < 0$

(D) $\Delta_rG^\circ(1) > 0; \Delta_rG^\circ(2) > 0$

Q19. MCQ (Single Correct Answer)

Open solution

Identify the CORRECT set of details from the following :

A. $[Co(NH_3)_6]^{3+}$: Inner orbital complex; d^2sp^3 hybridized

B. $[MnCl_6]^{3-}$: Outer orbital complex; $sp^3\,d^2$ hybridized

C. $[CoF_6]^{3-}$: Outer orbital complex; d^2sp^3 hybridized

D. $[FeF_6]^{3-}$: Outer orbital complex; $sp^3\,d^2$ hybridized

E. $[Ni(CN)_4]^{2-}$: Inner orbital complex; sp^3 hybridized

Choose the correct answer from the options given below :

(A) C & D Only **(B)** A, C & E Only **(C)** A, B, C, D & E **(D)** A, B & D Only

Q20. MCQ (Single Correct Answer)

Open solution

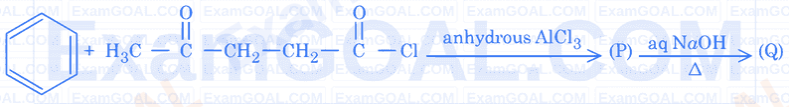
Elements X and Y belong to Group 15. The difference between the electronegativity values of ' X ' and phosphorus is higher than that of the difference between phosphorus and ' Y '. ' X ' & ' Y ' are respectively

(A) As & Bi **(B)** Bi & N **(C)** N & As **(D)** As & Sb

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q21. Numerical

Consider the following reaction of benzene.



In compound (Q), the percentage of oxygen is ____ %. (Nearest integer)

Open solution

Q22. Numerical

Total number of unpaired electrons present in the central metal atoms/ions of

$[Ni(CO)_4], [NiCl_4]^{2-}, [PtCl_2(NH_3)_2], [Ni(CN)_4]^{2-}$ and $[Pt(CN)_4]^{2-}$ is ____.

Open solution

Q23. Numerical

200 cc of $x \times 10^{-3}$M potassium dichromate is required to oxidise 750 cc of 0.6 M Mohr's salt solution in acidic medium.

Here $x =$ ____ .

Open solution

Q24. Numerical

$X_2(g) + Y_2(g) \rightleftharpoons 2Z(g)$

$X_2(g)$ and $Y_2(g)$ are added to a 1 L flask and it is found that the system attains the above equilibrium at $T(K)$ with the number of moles of $X_2(g), Y_2(g)$ and $Z(g)$ being 3,3 and 9 mol respectively (equilibrium moles). Under this condition of equilibrium, 10 mol of $Z(g)$ is added to the flask and the temperature is maintained at $T(K)$. Then the number of moles of $Z(g)$ in the flask when the new equilibrium is established is ____ . (Nearest integer)

Open solution

Q25. Numerical

Two liquids A and B form an ideal solution. At 320 K , the vapour pressure of the solution, containing 3 mol of A and 1 mol of B is 500 mm Hg . At the same temperature, if 1 mol of A is further added to this solution, vapour pressure of the solution increases by 20 mm Hg . Vapour pressure (in mm Hg) of B in pure state is ____ . (Nearest integer)

Open solution

Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

The number of ways, in which 16 oranges can be distributed to four children such that each child gets at least one orange, is

(A) 384 (B) 403 (C) 429 (D) 455

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q27. MCQ (Single Correct Answer)

Let $\sum_{k=1}^{n} a_k = \alpha n^2 + \beta n$. If $a_{10} = 59$ and $a_6 = 7a_1$, then $\alpha + \beta$ is equal to :

(A) 3 (B) 5 (C) 7 (D) 12

Open solution

Q28. MCQ (Single Correct Answer)

If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}, i = \sqrt{-1}$, then $\left(z^{201} - i\right)^8$ is equal to

(A) 1 (B) 0 (C) -1 (D) 256

Open solution

Q29. MCQ (Single Correct Answer)

Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $\vec{a} \times \vec{b} = 2(\vec{a} \times \vec{c})$. If $|\vec{a}| = 1, |\vec{b}| = 4, |\vec{c}| = 2$, and the angle between $\vec{b}$ and $\vec{c}$ is 60°, then $|\vec{a} \cdot \vec{c}|$ is equal to

(A) 2 (B) 0 (C) 1 (D) 4

Open solution

Q30. MCQ (Single Correct Answer)

Consider two sets $\mathrm{A} = \{x \in \mathrm{Z} : |(|x-3|-3)| \leq 1\}$ and

$\mathrm{B} = \left\{x \in \mathbb{R} - \{1,2\} : \frac{(x-2)(x-4)}{x-1} \log_e(|x-2|) = 0\right\}$. Then the number of

onto functions $f : \mathrm{A} \to \mathrm{B}$ is equal to :

(A) 32 (B) 81 (C) 79 (D) 62

Open solution

Q31. MCQ (Single Correct Answer)

Let $\mathrm{A} = \{0, 1, 2, \ldots, 9\}$. Let R be a relation on A defined by $(x, y) \in \mathrm{R}$ if and only if $|x - y|$ is a multiple of 3.

Given below are two statements :

Statement I : $n(\mathrm{R}) = 36$.

Statement II : R is an equivalence relation.

In the light of the above statements, choose the ***correct*** answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is incorrect but Statement II is correct

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q32. MCQ (Single Correct Answer)

Bag A contains 9 white and 8 black balls, while bag B contains 6 white and 4 black balls. One ball is randomly picked up from the bag B and mixed up with the balls in the bag A . Then a ball is randomly drawn from the bag A . If the probability, that the ball drawn is white, is $\frac{p}{q}, \gcd(p, q) = 1$, then $p + q$ is equal to

(A) 24 (B) 22 (C) 23 (D) 21

Open solution

Q33. MCQ (Single Correct Answer)

Let PQ be a chord of the hyperbola $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$, perpendicular to the x -axis such that OPQ is an equilateral triangle, O being the centre of the hyperbola. If the eccentricity of the hyperbola is $\sqrt{3}$, then the area of the triangle OPQ is

(A) $2\sqrt{3}$ (B) $\frac{11}{5}$ (C) $\frac{8\sqrt{3}}{5}$ (D) $\frac{9}{5}$

Open solution

Q34. MCQ (Single Correct Answer)

If the mean and the variance of the data

Class	$4-8$	$8-12$	$12-16$	$16-20$
Frequency	3	λ	4	7

are μ and 19 respectively, then the value of $\lambda + \mu$ is :

(A) 21 (B) 19 (C) 18 (D) 20

Open solution

Q35. MCQ (Single Correct Answer)

Let $A(1, 2)$ and $C(-3, -6)$ be two diagonally opposite vertices of a rhombus, whose sides AD and BC are parallel to the line $7x - y = 14$. If $B(\alpha, \beta)$ and $D(\gamma, \delta)$ are the other two vertices, then $|\alpha + \beta + \gamma + \delta|$ is equal to :

(A) 3 (B) 6 (C) 1 (D) 9

Open solution

Q36. MCQ (Single Correct Answer)

The sum of all the real solutions of the equation $\log_{(x+3)}\left(6x^2 + 28x + 30\right) = 5 - 2\log_{(6x+10)}\left(x^2 + 6x + 9\right)$ is equal to :

(A) 1 (B) 4 (C) 0 (D) 2

Open solution

Q37. MCQ (Single Correct Answer)

The least value of $\left(\cos^2\theta - 6\sin\theta\cos\theta + 3\sin^2\theta + 2\right)$ is

(A) $4 - \sqrt{10}$ (B) -1 (C) $4 + \sqrt{10}$ (D) 1

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q38. MCQ (Single Correct Answer)

Let $\frac{\pi}{2} < \theta < \pi$ and $\cot\theta = -\frac{1}{2\sqrt{2}}$. Then the value of

$\sin\left(\frac{15\theta}{2}\right)(\cos 8\theta + \sin 8\theta) + \cos\left(\frac{15\theta}{2}\right)(\cos 8\theta - \sin 8\theta)$

is equal to :

(A) $\frac{\sqrt{2}-1}{\sqrt{3}}$ (B) $\frac{\sqrt{2}}{\sqrt{3}}$ (C) $\frac{1-\sqrt{2}}{\sqrt{3}}$ (D) $-\frac{\sqrt{2}}{\sqrt{3}}$

Open solution

Q39. MCQ (Single Correct Answer)

Let $\mathrm{I}(x) = \int \frac{3dx}{(4x+6)\left(\sqrt{4x^2+8x+3}\right)}$ and $\mathrm{I}(0) = \frac{\sqrt{3}}{4} + 20$. If

$\mathrm{I}\left(\frac{1}{2}\right) = \frac{a\sqrt{2}}{b} + \mathrm{c}$, where $a, b, \mathrm{c} \in \mathrm{N}, \gcd(a,b) = 1$, then $a+b+c$ is equal to :

(A) 30 (B) 29 (C) 28 (D) 31

Open solution

Q40. MCQ (Single Correct Answer)

If the points of intersection of the ellipses $x^2 + 2y^2 - 6x - 12y + 23 = 0$ and

$4x^2 + 2y^2 - 20x - 12y + 35 = 0$ lie on a circle of radius r and centre (a, b), then the

value of $ab + 18r^2$ is :

(A) 53 (B) 52 (C) 55 (D) 51

Open solution

Q41. MCQ (Single Correct Answer)

Let $\vec{a} = \hat{\mathrm{i}} - 2\hat{\mathrm{j}} + 3\hat{\mathrm{k}}, \vec{b} = 2\hat{\mathrm{i}} + \hat{\mathrm{j}} - \hat{\mathrm{k}}, \vec{c} = \lambda\hat{\mathrm{i}} + \hat{\mathrm{j}} + \hat{\mathrm{k}}$ and $\vec{v} = \vec{a} \times \vec{b}$. If $\vec{v} \cdot \vec{c} = 11$ and the length of the projection of $\vec{b}$ on $\vec{c}$ is p, then $9p^2$ is equal to

(A) 9 (B) 6 (C) 4 (D) 12

Open solution

Q42. MCQ (Single Correct Answer)

If $f(x) = \begin{cases} \frac{a|x| + x^2 - 2(\sin|x|)(\cos|x|)}{x} & , x \neq 0 \\ b & , x = 0 \end{cases}$

is continuous at $x = 0$, then $a + b$ is equal to :

(A) 1 (B) 2 (C) 0 (D) 4

Open solution

Q43. MCQ (Single Correct Answer)

The area of the region enclosed between the circles $x^2 + y^2 = 4$ and $x^2 + (y-2)^2 = 4$ is:

(A) $\frac{2}{3}(4\pi - 3\sqrt{3})$ (B) $\frac{4}{3}(2\pi - \sqrt{3})$ (C) $\frac{4}{3}(2\pi - 3\sqrt{3})$ (D) $\frac{2}{3}(2\pi - 3\sqrt{3})$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q44. MCQ (Single Correct Answer)

An equilateral triangle OAB is inscribed in the parabola $y^2 = 4x$ with the vertex O at the vertex of the parabola. Then the minimum distance of the circle having AB as a diameter from the origin is

(A) $2(8 - 3\sqrt{3})$ (B) $4(6 + \sqrt{3})$ (C) $4(3 - \sqrt{3})$ (D) $2(3 + \sqrt{3})$

Open solution

Q45. MCQ (Single Correct Answer)

The system of linear equations

$x + y + z = 6$
$2x + 5y + az = 36$
$x + 2y + 3z = b$

has :

(A) unique solution for $a = 8$ and $b = 16$

(B) infinitely many solutions for $a = 8$ and $b = 14$

(C) infinitely many solutions for $a = 8$ and $b = 16$

(D) unique solution for $a = 8$ and $b = 14$

Open solution

Q46. Numerical

If the image of the point $\mathrm{P}(a, 2, a)$ in the line $\frac{x}{2} = \frac{y+a}{1} = \frac{z}{1}$ is Q and the image of Q in the line $\frac{x-2b}{2} = \frac{y-a}{1} = \frac{z+2b}{-5}$ is P , then $a + b$ is equal to ____.

Open solution

Q47. Numerical

If the solution curve $y = f(x)$ of the differential equation $(x^2 - 4)y' - 2xy + 2x(4 - x^2)^2 = 0, x > 2,$ passes through the point $(3, 15)$, then the local maximum value of f is ____

Open solution

Q48. Numerical

The number of elements in the set $\mathrm{S} = \{x : x \in [0, 100] \text{ and } \int_0^x t^2 \sin(x - t)\mathrm{d}t = x^2\}$ is ____

Open solution

Q49. Numerical

Let S denote the set of 4-digit numbers $abcd$ such that $a > b > c > d$ and P denote the set of 5 -digit numbers having product of its digits equal to 20 . Then $n(\mathrm{S}) + n(\mathrm{P})$ is equal to ____

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q50. Numerical

Let $A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{bmatrix}$ and B be a matrix such that $B(I - A) = I + A$. Then the sum of the diagonal elements of $\mathrm{B^TB}$ is equal to ____

Open solution

Physics 25 questions

Q51. MCQ (Single Correct Answer)

One mole of an ideal diatomic gas expands from volume V to $2V$ isothermally at a temperature $27^\circ\mathrm{C}$ and does W joule of work. If the gas undergoes same magnitude of expansion adiabatically from $27^\circ\mathrm{C}$ doing the same amount of work W, then its final temperature will be (close to) ____ $^\circ\mathrm{C}$.

$(\log_e 2 = 0.693)$

(A) -56 (B) -117 (C) -30 (D) -189

Open solution

Q52. MCQ (Single Correct Answer)

A block is sliding down on an inclined plane of slope θ and at an instant $t = 0$ this block is given an upward momentum so that it starts moving up on the inclined surface with velocity u. The distance (S) travelled by the block before its velocity become zero, is ____ .

(g = gravitational acceleration)

(A) $\frac{2u^2}{\mathrm{g}\cos\theta}$ (B) $\frac{u^2}{\sqrt{2}g\cos\theta}$

(C) $\frac{u^2}{2g\sin\theta}$ (D) $\frac{u^2}{2g\cos\theta}$

Open solution

Q53. MCQ (Single Correct Answer)

The internal energy of a monoatomic gas is 3nRT. One mole of helium is kept in a cylinder having internal cross section area of $17\ \mathrm{cm}^2$ and fitted with a light movable frictionless piston. The gas is heated slowly by suppling 126 J heat. If the temperature rises by $4^\circ\mathrm{C}$, then the piston will move ____ cm.

(atmospheric pressure $= 10^5\ \mathrm{Pa}$)

(A) 1.55 (B) 14.5 (C) 15.5 (D) 1.45

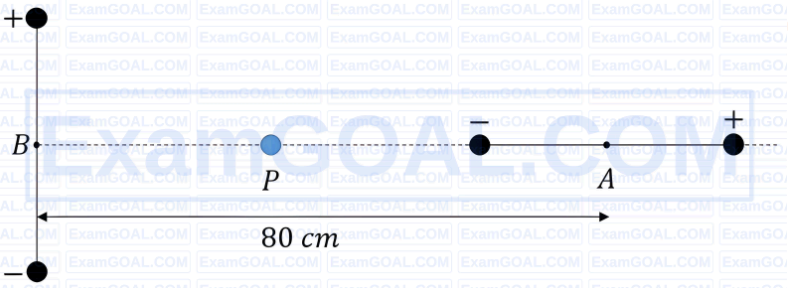
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q54. MCQ (Single Correct Answer)

Open solution

Two short dipoles (A, B), A having charges $\pm 2\mu C$ and length 1 cm and B having charges $\pm 4\mu C$ and length 1 cm are placed with their centres 80 cm apart as shown in the figure. The electric field at a point P, equi-distant from the centres of both dipoles is ____ N/C.



(A) $\frac{9}{16}\sqrt{2} \times 10^5$

(B) $\frac{9}{16}\sqrt{2} \times 10^4$

(C) $9\sqrt{2} \times 10^4$

(D) $4.5\sqrt{2} \times 10^4$

Q55. MCQ (Single Correct Answer)

Open solution

A circular loop of radius 7 cm is placed in uniform magnetic field of 0.2 T directed perpendicular to plane of loop. The loop is converted into a square loop in 0.5 s . The EMF induced in the loop is ____ mV.

(A) 13.2

(B) 6.6

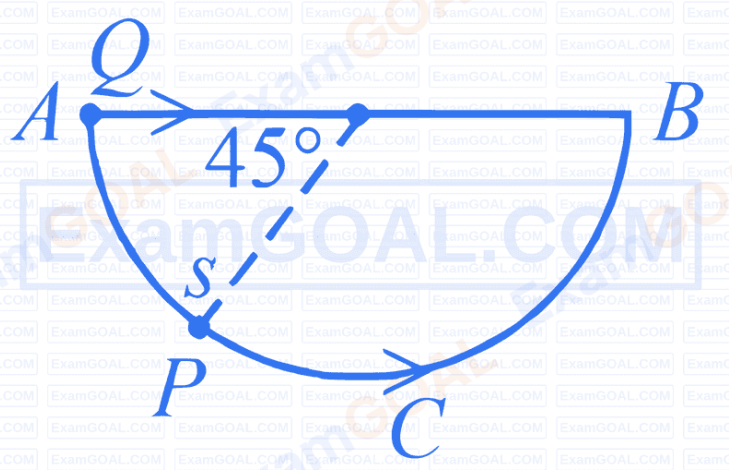
(C) 1.32

(D) 8.25

Q56. MCQ (Single Correct Answer)

Open solution

A bead P sliding on a frictionless semi-circular string (ACB) and it is at point S at $t = 0$ and at this instant the horizontal component of its velocity is v. Another bead Q of the same mass as P is ejected from point A at $t = 0$ along the horizontal string AB, with the speed v, friction between the beads and the respective strings may be neglected in both cases. Let t_P and t_Q be the respective times taken by beads P and Q to reach the point B, then the relation between t_P and t_Q is



(A) $t_P < t_Q$

(B) $t_P = t_Q$

(C) $t_P > 1.25t_Q$

(D) $t_P > t_Q$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q57. MCQ (Single Correct Answer)

A paratrooper jumps from an aeroplane and opens a parachute after 2 s of free fall and starts decelerating with $3\ \mathrm{m/s^2}$. At 10 m height from ground, while descending with the help of parachute, the speed of paratrooper is $5\ \mathrm{m/s}$. The initial height of the airplane is ____ m.

$\left(g = 10\ \mathrm{m/s^2}\right)$

(A) 92.5 (B) 62.5 (C) 20 (D) 82.5

Open solution

Q58. MCQ (Single Correct Answer)

The ratio of speeds of electromagnetic waves in vacuum and a medium, having dielectric constant $k = 3$ and permeability of $\mu = 2\mu_0$, is ($\mu_0 =$ permeability of vacuum)

(A) $6:1$ (B) $3:2$ (C) $\sqrt{6}:1$ (D) $36:1$

Open solution

Q59. MCQ (Single Correct Answer)

Suppose a long solenoid of 100 cm length, radius 2 cm having 500 turns per unit length, carries a current $I = 10\sin(\omega t)\mathrm{A}$, where $\omega = 1000\mathrm{rad.\ /s}$. A circular conducting loop (B) of radius 1 cm coaxially slided through the solenoid at a speed $v = 1\ \mathrm{cm/s}$. The r.m.s. current through the loop when the coil B is inserted 10 cm inside the solenoid is $\frac{\alpha}{\sqrt{2}}\mu A$. The value of α is ____ .

[Resistance of the loop $= 10\Omega$]

(A) 80 (B) 280 (C) 100 (D) 197

Open solution

Q60. MCQ (Single Correct Answer)

A body of mass 14 kg initially at rest explodes and breaks into three fragments of masses in the ratio $2:2:3$. The two pieces of equal masses fly off perpendicular to each other with a speed of $18\ \mathrm{m/s}$ each. The velocity of the heavier fragment is

____ $\mathrm{m/s}$.

(A) $24\sqrt{2}$ (B) 12 (C) $12\sqrt{2}$ (D) $10\sqrt{2}$

Open solution

Q61. MCQ (Single Correct Answer)

A parallel plate capacitor with plate separation 5 mm is charged by a battery. On introducing a mica sheet of 2 mm and maintaining the connections of the plates with the terminals of the battery, it is found that it draws 25% more charge from the battery. The dielectric constant of mica is ____

(A) 1.0 (B) 2.5 (C) 1.5 (D) 2.0

Open solution

Q62. MCQ (Single Correct Answer)

Which of the following pair of nuclei are isobars of the element?

(A) ${}_1^3\mathrm{H}$ and ${}_2^3\mathrm{He}$ (B) ${}_{80}^{198}\mathrm{Hg}$ and ${}_{79}^{197}\mathrm{Au}$

(C) ${}_1^2\mathrm{H}$ and ${}_1^3\mathrm{H}$ (D) ${}_{92}^{236}\mathrm{U}$ and ${}_{92}^{238}\mathrm{U}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q63. MCQ (Single Correct Answer)

To compare EMF of two cells using potentiometer the balancing lengths obtained are 200 cm and 150 cm . The least count of scale is 1 cm . The percentage error in the ratio of EMFs is ____

(A) 1.75 (B) 1.55 (C) 1.17 (D) 1.65

Open solution

Q64. MCQ (Single Correct Answer)

Two charges $7\mu C$ and $-2\mu C$ are placed at $(-9,0,0)\text{cm}$ and $(9,0,0)\text{cm}$ respectively in an external field $E=\frac{A}{r^2}\hat{r}$, where $A=9\times10^5\ \text{N/C.m}^2$. Considering the potential at infinity is 0 , the electrostatic energy of the configuration is ____ J.

(A) 49.3 (B) 1.4 (C) 24.3 (D) -90.7

Open solution

Q65. MCQ (More than One Correct Answer)

A prism of angle 75° and refractive index $\sqrt{3}$ is coated with thin film of refractive index 1.5 only at the back exit surface. To have total internal reflection at the back exit surface the incident angle must be ____

$(\sin 15^\circ = 0.25$ and $\sin 25^\circ = 0.43)$

(A) between 15° and 20° (B) 15°

(C) $< 15^\circ$ (D) $> 25^\circ$

Open solution

Q66. MCQ (Single Correct Answer)

The current passing through a conducting loop in the form of equilateral triangle of side $4\sqrt{3}\ \text{cm}$ is 2 A . The magnetic field at its centroid is $\alpha\times10^{-5}\ \text{T}$. The value of α is ____ .

(Given : $\mu_o = 4\pi\times10^{-7}$ SI units)

(A) $3\sqrt{3}$ (B) $2\sqrt{3}$ (C) $\sqrt{3}$ (D) $\frac{\sqrt{3}}{2}$

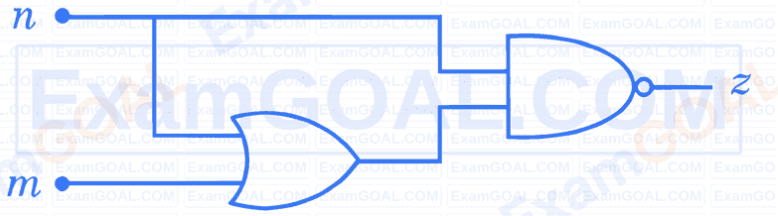
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q67. MCQ (Single Correct Answer)

Open solution

For the given logic gate circuit, which of the following is the correct truth table?



(A)

n	m	z
0	0	1
0	1	1
1	1	0
1	0	0

(B)

n	m	z
0	0	1
0	1	0
1	1	1
1	0	0

(C)

n	m	z
0	0	0
0	1	1
1	1	0
1	0	1

(D)

n	m	z
0	0	1
0	1	0
1	1	0
1	0	0

Q68. MCQ (Single Correct Answer)

Open solution

A small metallic sphere of diameter 2 mm and density $10.5\ \mathrm{g/cm^3}$ is dropped in glycerine having viscosity 10 Poise and density $1.5\ \mathrm{g/cm^3}$ respectively. The terminal velocity attained by the sphere is ____ $\mathrm{cm/s}$.

$\left(\pi = \frac{22}{7} \text{ and } g = 10\ \mathrm{m/s^2}\right)$

(A) 1.0 (B) 1.5 (C) 3.0 (D) 2.0

Q69. MCQ (Single Correct Answer)

Open solution

An air bubble of volume $2.9\ \mathrm{cm^3}$ rises from the bottom of a swimming pool of 5 m deep. At the bottom of the pool water temperature is $17^\circ\mathrm{C}$. The volume of the bubble when it reaches the surface, where the water temperature is $27^\circ\mathrm{C}$, is ____ $\mathrm{cm^3}$.

($\mathrm{g} = 10\ \mathrm{m/s^2}$, density of water $= 10^3\ \mathrm{kg/m^3}$, and 1 atm pressure is $10^5\ \mathrm{Pa}$)

(A) 2.0 (B) 4.2 (C) 3.0 (D) 4.5

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q70. MCQ (Single Correct Answer)

When an unpolarized light falls at a particular angle on a glass plate (placed in air), it is observed that the reflected beam is linearly polarized. The angle of refracted beam with respect to the normal is ____ .

$(\tan^{-1}(1.52) = 57.7^\circ$, refractive indices of air and glass are 1.00 and 1.52, respectively.)

(A) 36.3° (B) 39.6° (C) 42.6° (D) 32.3°

Open solution

Q71. Numerical

The size of the images of an object, formed by a thin lens are equal when the object is placed at two different positions 8 cm and 24 cm from the lens. The focal length of the lens is ____ cm.

Open solution

Q72. Numerical

The average energy released per fission for the nucleus of $^{235}_{92}\mathrm{U}$ is 190 MeV . When all the atoms of 47 g pure $^{235}_{92}\mathrm{U}$ undergo fission process, the energy released is $\alpha \times 10^{23}\mathrm{MeV}$. The value of α is ____ .

(Avogadro Number $= 6 \times 10^{23}$ per mole)

Open solution

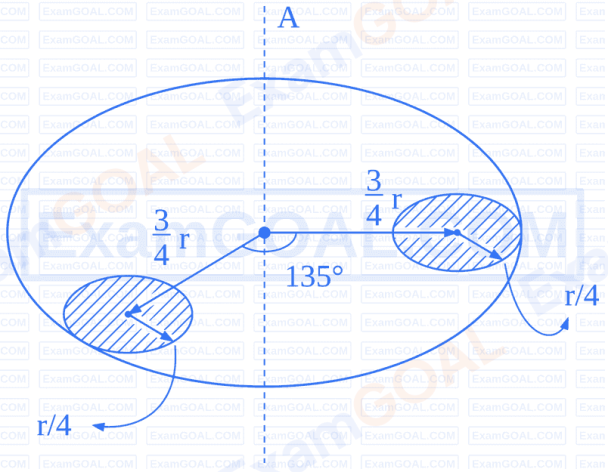
Q73. Numerical

The velocity of sound in air is doubled when the temperature is raised from $0^\circ\mathrm{C}$ to $\alpha^\circ\mathrm{C}$. The value of α is ____ .

Open solution

Q74. Numerical

Suppose there is a uniform circular disc of mass M kg and radius r m shown in figure. The shaded regions are cut out from the disc. The moment of inertia of the remainder about the axis A of the disc is given by $\frac{x}{256}Mr^2$. The value of x is ____ .



Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q75. Numerical

A ball of radius r and density ρ dropped through a viscous liquid of density σ and viscosity η attains its terminal velocity at time t, given by $t = A\rho^a r^b \eta^c \sigma^d$, where A is a constant and a, b, c and d are integers. The value of $\frac{b+c}{a+d}$ is ____ .

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Chemistry

1. **D**	2. **B**	3. **B**	4. **D**	5. **D**
6. **A**	7. **A**	8. **C**	9. **C**	10. **B**
11. **B**	12. **B**	13. **C**	14. **B**	15. **D**
16. **B**	17. **C**	18. **B**	19. **D**	20. **C**
21. **10**	22. **2**	23. **375**	24. **15**	25. **200**

Mathematics

26. **D**	27. **B**	28. **D**	29. **C**	30. **D**
31. **D**	32. **C**	33. **C**	34. **B**	35. **B**
36. **C**	37. **A**	38. **C**	39. **D**	40. **C**
41. **D**	42. **B**	43. **A**	44. **C**	45. **B**
46. **3**	47. **16**	48. **16**	49. **260**	50. **3**

Physics

51. **A**	52. **C**	53. **C**	54. **B**	55. **C**
56. **A**	57. **A**	58. **C**	59. **D**	60. **C**
61. **D**	62. **A**	63. **C**	64. **A**	65. **A, B, C**
66. **A**	67. **A**	68. **D**	69. **D**	70. **D**
71. **16**	72. **228**	73. **819**	74. **109**	75. **1**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs